

SHOW ME THE NUMBERS

Designing Tables and Graphs to Enlighten

SECOND EDITION

STEPHEN FEW

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5 VISUAL PERCEPTION AND GRAPHICAL COMMUNICATION

Because graphical communication is visual, it must express information in ways that human eyes can perceive and brains can understand. Our eyes and the parts of the brain that handle input from them work in particular ways. Thanks to science, how we see is now fairly well understood, from the initial information-carrying rays of light that enter our eyes to the interpretation of that information in the gray folds of the visual cortex. By understanding visual perception and its application to the graphical communication of quantitative information, you will learn what works, what doesn't, and why. This chapter brings the principles of graphical design for communication alive in ways that are practical and can be applied skillfully to real-world challenges in presenting quantitative information.

Vision, of all the senses, is our most powerful and efficient channel for receiving information from the world around us. Approximately 70% of the sense receptors in our bodies are dedicated to vision.

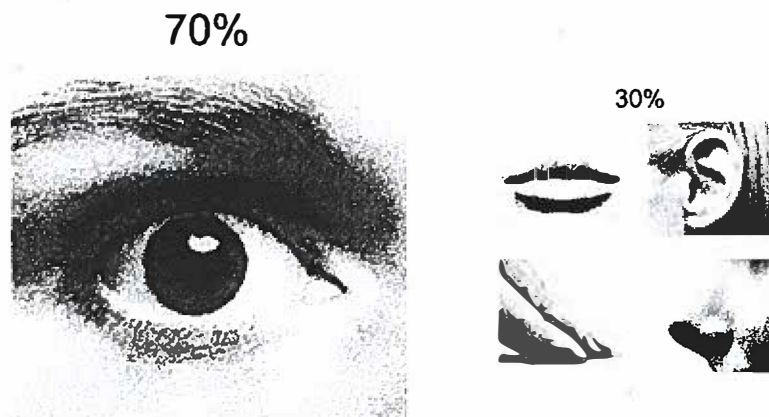


FIGURE 5.1 70% of the body's sense receptors reside in the retinas of our eyes.

There is an intimate connection among seeing, thinking, and understanding. It is no accident that the terms we most often use to describe understanding are visual in nature, such as insight, illumination, and enlightenment. "I see" is the expression we use when something begins to make sense. We call people of extraordinary wisdom seers.

Graphs, and, to a lesser extent, tables, are visual means of communication. Although tables are visual in that we receive their information through our eyes, their reliance on verbal language in the form of written text reduces the degree to which they take advantage of the power of visual perception. Nevertheless, inattention to the visual design of tables, such as improper alignment of numbers and excessive use of lines and fill colors, can greatly diminish their effectiveness. Graphs, in contrast, are primarily visual in nature. The power of graphs and the power of visual perception are intimately linked.

Equipped with a fundamental understanding of visual perception—how it works, what works, what doesn't, and why—your understanding of table and graph design will deepen beyond a memorized list of design principles. You will be able to apply these insights to new design challenges that extend well beyond the examples found in this book.

Colin Ware has written two wonderful books that explain how visual perception works and how this knowledge can be applied to graphical communication. Here's how he expresses the significance of visual perception:

Why should we be interested in visualization? . . . The visual system has its own rules. We can easily see patterns presented in certain ways, but if they are presented in other ways, they become invisible . . . When data is presented in certain ways, the patterns can be readily perceived. If we can understand how perception works, our knowledge can be translated into rules for displaying information. Following perception-based rules, we can present our data in such a way that the important and informative patterns stand out. If we disobey the rules, our data will be incomprehensible or misleading.¹

The rules that the visual system follows are often not obvious. In the following figure, one set of circles appears to be bulging outwards and the other appears to be dented inwards. Which is bulging outwards?



The one on the left—correct? Now I'm going to make a change and ask the question again. Which set is bulging outwards now?



Now the circles on the right appear to be bulging outwards, but I did not switch the position of the two sets; I merely turned the circles upside down. There is a reason why those on the right appear to bulge outwards: built into visual perception is the assumption that the light is coming from above. Because it looks as if light is reflecting off of their upper halves and shadows appear on

1. Colin Ware (2004) *Information Visualization: Perception for Design*, Second Edition. Morgan Kaufmann Publishers, page xxi.

FIGURE 5.2 One set of circles appears to be bulging outwards and the other is dented inwards.

FIGURE 5.3 The set that appears to be bulging outwards has now switched positions.

their lower halves, the circles on the right are naturally perceived as bulging outwards. Until the invention of fire and more recently the invention of electric lights, all significant sources of illumination came from the sun, so light almost always came from above. Now that other sources of light are commonplace, perhaps this assumption will eventually disappear from the wiring in our brains, but for now it guides our perceptions. I mention this only to point out that rules are built into visual perception that aren't always obvious. We need to understand these rules if we want to present data in ways that work for people's eyes and make sense to their brains.

Mechanics of Sight

The process of seeing and making sense of what we see begins with sensation in our eyes and progresses to perception in our brains. Sensation is a physical process that occurs when a stimulus is detected. Perception is a cognitive process, involving the interpretation of the physical stimulus to make sense of it. For a visual stimulus to occur, there must be light. To see something, there must be an object out there that is either itself a source of light (e.g., the sun, a light bulb, a computer screen, a firefly) or a surface that reflects light (e.g., an apple, a piece of paper on which a table or graph is printed).

The light enters our eyes, which contain nerve cells specifically designed to absorb light and translate it into neural signals (the language of the nervous system). These neural signals are then passed via the optic nerve to our brains where they are processed in a series of stages to make sense out of what we've seen.

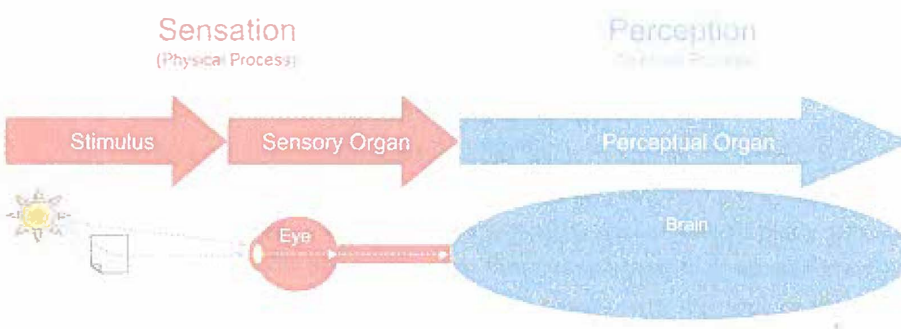


FIGURE 5.4 This diagram represents the stages in the process of visual perception.

The Eye

Our eyes operate a lot like cameras, with several components working together to sense information that is contained in light. The surface of the eye, called the *cornea*, is a protective covering. Inside the eye, behind the cornea, resides the *iris*, a muscle that works like a camera shutter, enlarging or decreasing the size of the opening, called the *pupil*, to allow just the right amount of light to enter. The pupil becomes small when the amount of light is high (i.e., bright) and enlarges when there is less light, thereby allowing enough light through to provide a discernable image. Behind the iris is a lens, similar to the lens of a camera, which thickens to focus on objects that are near and thins to focus on distant objects.

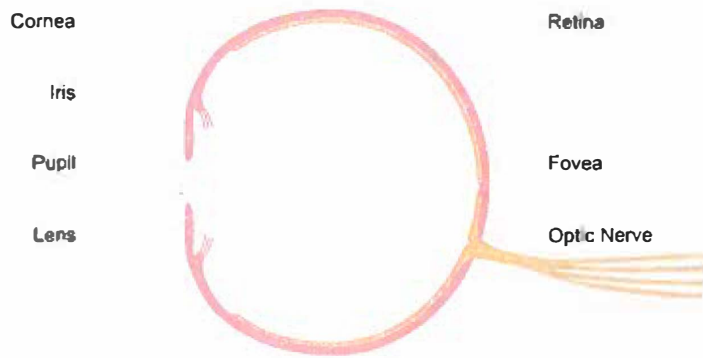


FIGURE 5.5 This is a diagram of the eye, with a label for each of the components that are significant to vision.

Eye diagram created by Keith Stevenson.

Once light passes through the pupil, it shines on the rear inner surface of the eye, called the *retina*. This is a thin surface coated with millions of specialized nerve cells called *rods* and *cones*. Rods and cones differ in photosensitivity: rods specialize in sensing dim light and record what they detect in black and white, and cones specialize in sensing brighter light and record what they detect in color. Cones are divided into three types, each of which specializes in detecting a different range of the color spectrum: blue, green, or red.

In the center of the retina is a small area called the *fovea* where cones are densely concentrated. Images that are focused on the fovea are seen with extreme clarity because of the high number of cones there. When we want to examine something closely, we focus on it directly so that the light coming from it is directed to the fovea. When we focus in this manner on a printed page, we are able to see an amazing amount of detail, making distinctions between as many as 625 separate points of ink per square inch.²

Rods and cones encode what they detect into neural signals: electro-chemical pulses, the language of the nervous system. These signals are then passed along the optic nerve to the brain where perceptual processing occurs. Millions of cones and rods work together simultaneously, each sending signals to the brain at the same time through an extremely high-bandwidth channel at amazing speeds. Visual perception is the most highly developed and efficient of our senses.

The process of seeing is not smooth as you might imagine but consists of spurts of information interrupted by frequent, brief pauses. This is because the fovea can only focus on a limited area at a time. Our eyes fixate on a particular spot and remain there for up to half a second, taking in the details while at the same time we use what we are seeing in the non-foveal parts of the retina to survey the full field of vision and select our next point of concentration. It takes our eyes about a quarter second to jump to the next point of fixation where the process is repeated. These little jumps are called *saccades*, and the motion is called *saccadic eye movement*.³

As you are reading this page, your eyes are quickly jumping and fixating, jumping and fixating, taking in a few words at a time, moving from left to right across each line, then down to the next. During the brief moment when a saccade is occurring, when your eyes are jumping from one point of fixation to the next, you are not as sensitive to visual sensation as you are when your eyes

2. Edward Tufte (2001) *The Visual Display of Quantitative Information*, Second Edition. Graphics Press, page 161.

3. Much of the information about the saccadic nature of vision was derived from Marlana Coe (1996) *Human Factors for Technical Communicators*. John Wiley & Sons, Inc.

are not moving. If something popped up on your computer screen during a saccadic eye movement and disappeared before the movement was complete, you would probably never notice that it was ever there.

Despite the millions of bits of information that our eyes process every second, we only see a little of what is out there, and only a fraction of that in detail. Some things grab our attention more than others.

The Brain

We don't see images with our eyes; we see them with our brains. Our eyes record light and translate it into electro-chemical signals, passing them on to our brains, which is where images are actually perceived. What goes on in the brain produces what we experience as visual perception.

Although the analogy is less than perfect, it is useful to think of the brain as a computer. Computers involve components and processes that are like the components and processes of our brains. Computers receive input from a variety of devices (a keyboard, mouse, digital camera, etc.). Let's trace the path of input from the keyboard as an example of how computers process input from the outside world, which is comparable to the sensory information we perceive. When we tap the keys of a keyboard, electrical signals are initially routed into a special kind of memory called a *buffer*, in this case the keyboard buffer. This temporary location serves as a waiting room for the fraction of a second that the computer is busy until it is ready for fresh input. Once the computer is ready for the next input, these electrical signals move along the path into *working memory*, also called *random access memory (RAM)*. This is an extremely high-speed form of memory that the computer uses to process the information, doing what is necessary to interpret the message and respond to it. In RAM, bits and bytes are translated into meaningful terms, such as commands to take a particular action or words of a document. If the information that the computer has received is meant to be kept around for future reference, it is then moved into some form of permanent memory, such as a hard disk. These three types of memory—buffer memory, working memory, and permanent memory—are analogous to three types of memory that process visual information in our brains, which are called:

- *Iconic memory*
- *Working memory*
- *Long-term memory*

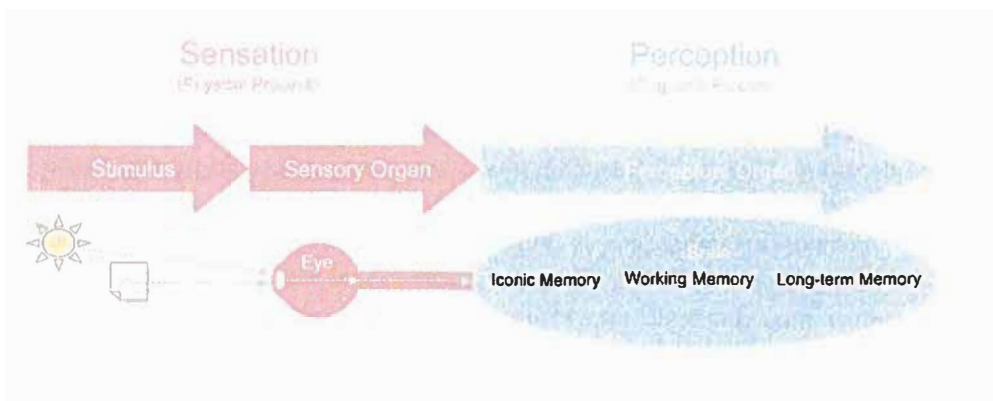


FIGURE 5.6 This is a diagram of visual perception, including the three types of memory that handle visual processing in the brain: iconic, working, and long-term memory.

ICONIC MEMORY

Signals from the eyes are initially routed through the optic nerve into iconic memory (a.k.a. the *visual sensory register*) where each snapshot of input waits to be passed on to working memory. Information ordinarily remains in iconic memory for less than a second while extremely rapid processing takes place before the information is forwarded to working memory. This part of perceptual processing is automatic and unconscious. For this reason it is called *preattentive processing*, as opposed to the conscious, higher-level cognitive processing that occurs later in working memory, which is called *attentive processing*. Preattentive processing is an extremely fast process of recognition and nothing more. Attentive processing, on the other hand, is a sequential process that takes more time and can result in learning, understanding, and remembering. Preattentive visual processing detects several attributes, such as color and the location of objects in 2-D space. Because preattentive processing is tuned to these attributes, they jump out at us and are therefore extremely powerful aspects of visual perception. If you want something to stand out as important in a table or graph, you should encode it using a preattentive attribute that contrasts with the surrounding information, such as red text in the midst of black text. If you want a particular set of objects to be seen as a group, you can do this by assigning them the same preattentive attribute.

Preattentive attributes play an important role in visual design. They can be used to distinctly group and highlight objects. We'll examine them in detail later in this chapter with special attention to how we can use them to design tables and graphs that communicate effectively.

WORKING MEMORY

As visual information is moved from iconic memory into working memory, what our brains deem useful is combined into meaningful visual *chunks*. These chunks of visual information are used by the conscious, attentive process of perception that occurs in working memory. Two fundamental characteristics of working memory are:

- It is temporary.
- It has limited storage capacity.

Information remains in working memory from only a few seconds to as long as a few hours if it is periodically rehearsed. Without rehearsal, it soon vanishes. With the right kind of rehearsal, it is stored in long-term memory where it remains for later use.

Only three or four chunks of information can be stored in working memory at any one time. For something new to be brought into working memory, something that is already there must either be moved into long-term memory or forgotten. Similar to RAM in a computer, working memory in the brain stores information temporarily for high-speed processing. As this processing occurs, new sensations or memories are being moved into working memory while others are either moved into long-term memory or forgotten.

Colin Ware points out, in *Information Visualization: Perception for Design*, that our brains appear to have more than one type of working memory; each kind specializes in a different type of information processing. For instance, research has demonstrated that there are separate areas for visual and verbal information and that these areas don't compete with each other for resources.

An important fact to keep in mind about working memory is that readers of tables and graphs can only hold a few chunks of information in their heads at any one time. For instance, if you design a graph that includes a legend with a different color or symbol shape for 10 different sets of data, your readers will be forced to constantly refer back to the legend to remind themselves which is which because working memory is limited to three or four concurrent chunks of information. Chunks of information can vary in size. By designing a visual display of information to form larger, coherent patterns that combine multiple data values into chunks, you can make it easier for your readers to hold more information in working memory. This is one of the reasons that graphs are capable of communicating a great deal of information that can be perceived all at once while tables are limited to use for looking things up. We can't take a bunch of numbers from a table and chunk them together meaningfully for storage in working memory; we can, however, discern in a graph the image of a single, meaningful pattern that consists of thousands of values.

LONG-TERM MEMORY

When we decide to store information for later use, either consciously or unconsciously, we rehearse that chunk of information to move it from working to long-term memory. How we store that information involves an intricate neural network of links and cross-references (i.e., associations), like indexes in a computer that help us to find information and retrieve it from a hard disk into working memory when we need it. One piece of information can have many associations.

Long-term memory is vitally important to visual perception because it holds our ability to recognize images and detect meaningful patterns, but we don't have to understand very much about long-term memory to become better designers of tables and graphs. Mostly, we need to know how to use preattentive visual attributes to grab and direct our readers' attention and how to work within the limits of working memory.

Evolution of Visual Perception

Like all other products of evolution, visual perception developed as an aid to survival. Because of its evolutionary roots, visual perception is fundamentally oriented toward action, always looking for what we can do with the things we see. As designers of tables and graphs, we should understand how to use attributes of visual perception to encode information in ways that others can easily, accurately, and efficiently decode.

Attributes of Preattentive Processing

Preattentive processing occurs below the level of consciousness at an extremely high speed and is tuned to detect a specific set of visual attributes. Attentive processing is conscious, sequential, and much slower. The difference between

preattentive processing and attentive processing is easy to demonstrate. Take a look at the four rows of numbers below and determine, as quickly as you can, how many times the number 5 appears in the list:



FIGURE 5.7 This example demonstrates the slow speed at which we perceive information using a serial process, one number at a time.

How many? The answer is six. Even if you got the right answer (many don't when I do this exercise in class), it took you several seconds to perform this task because it involved attentive processing. The list of numbers did not include any preattentive attributes that could be used to distinguish the fives from the other numbers, so you were forced to perform a sequential search looking for the specific shape of the number five. Now do it again, this time using the list below:

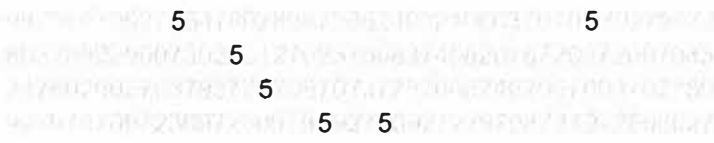


FIGURE 5.8 This example demonstrates the fast speed with which we process visual stimuli that exhibit preattentive attributes.

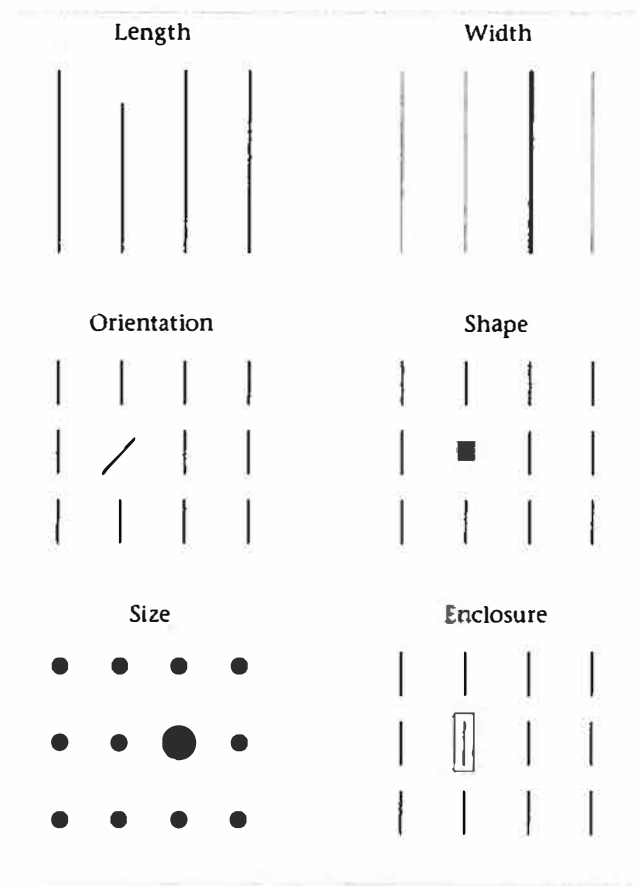
This time it was easy because the fives were distinguished through the preattentive attribute of color intensity. Because only the fives are black, and all the other numbers are light gray, the black fives stand out in contrast to the rest. This example shows the power of preattentive attributes used knowledgeably for visual communication.

Colin Ware has organized preattentive attributes into four categories: form, color, spatial position, and motion. I've reduced his list of attributes somewhat and removed the category of motion entirely, to focus on attributes that we find most relevant to static tables and graphs:

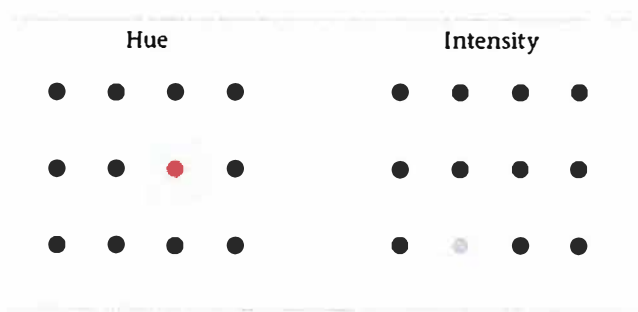
Colin Ware (2004) *Information Visualization: Perception for Design*, Second Edition. Morgan Kaufmann Publishers, pages 151 and 152

Category	Attribute
Form	Length
	Width
	Orientation
	Shape
	Size
	Enclosure
Color	Hue
	Intensity
Spatial position	2-D position

In each illustration of a preattentive attribute on the following page, only one object stands out as different from the rest, based on a variation of the attribute.

Attributes of Form

For our purposes, length is distance along the dominant dimension of an object. For example, the lines in the illustration above primarily vary in the vertical dimension only. Think in terms of bars in a bar graph, which encode quantitative values as length along a single dimension—either vertically or horizontally, but not both. Whenever we use variation in length to encode values in graphs, the objects, such as bars, will share a common baseline, which makes it extremely easy to compare their lengths. By width, we're referring to variation along the secondary dimension of an object. Once again, think of bars that vary in one dimension, such as vertical length, to encode values, but also vary in their less dominant dimension, width, to encode a second set of values.

Attributes of Color

Hue is a precise term for what we normally think of as color (red, green, blue, pink, etc.). Color is made up of three separate attributes; hue is one. *Intensity* applies to both of the other two attributes of color: *saturation* and *lightness*. In the previous illustration only intensity varies in that one object is lighter than the others.

The primary system that is used to describe color is known as the *HSL* (hue, saturation, and lightness) system. This system describes hue numerically in degrees from 0 to 360 along the circumference of a circle, illustrated by the following color wheel:

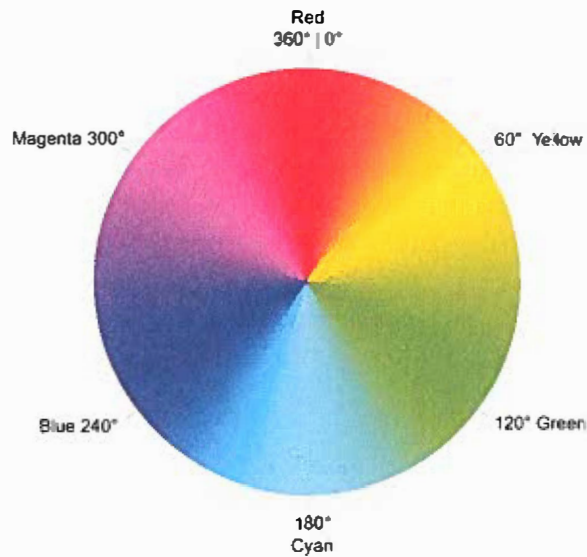


FIGURE 5.9 This is a color wheel, which represents hue in a circular manner.

Numbering starts at the top with red at 0° and continues around the wheel until it reaches the point where it started with red again, which is also assigned the value of 360°. If you could walk around this color wheel clockwise taking 60° steps, starting at red (0°), with each succeeding step you would be positioned at the following hues: yellow (60°), green (120°), cyan (180°), blue (240°), magenta (300°), then back to red again (360°).

Saturation measures the degree to which a particular hue fully exhibits its essence. In the example below, red ranges from completely pale on the left, lacking the essence of red entirely, to fully red on the right, exhibiting its full essence. Saturation is expressed as a percentage, starting with 0% on the left, which represents no saturation, to 100% on the right, which represents full saturation.



FIGURE 5.10 This is an illustration of the full range of saturation for the hue red, with 0% saturation on the left and 100% saturation on the right.

Lightness (also known as brightness) measures the degree to which a color appears dark or bright. In the next example, red ranges from fully dark (black) on the left to bright red on the right. Lightness is also expressed as a percentage, with 0% representing fully dark and 100% representing fully light.



FIGURE 5.11 This is an illustration of the full range of lightness for the hue red, with 0% lightness (dark) on the left and 100% lightness (bright) on the right.

Any color can be described numerically based on these three measures: hue (0 – 360°), saturation (0 – 100%) and lightness (0 – 100%).

Attributes of Spatial Position

2-D Position



Our perception of space is primarily two dimensional. We perceive differences in vertical position (up and down) and in horizontal position (left and right) clearly and accurately. We also perceive a third dimension, depth, but not nearly as well.

Applying Visual Attributes to Design

Preattentive attributes can be used to make particular aspects of what we see stand out from the rest. Understanding these attributes enables us to design tables and graphs that visually emphasize the most important information they contain, keeping the other visual elements from competing for attention with key information.

Uses for Encoding Quantitative Values

Some attributes of preattentive processing can be expressed as values along a continuum. For instance, the preattentive attribute of length can be expressed as a line of any length that the eye can see, not just simply as short and long. We perceive quantitative meanings in the variations of some visual attributes but not in others. Using the attribute of length once again as an example, we naturally perceive long lines as greater than short lines. In this sense, we perceive the differing values of length quantitatively. But what about an attribute like hue? Which represents the greater value: blue, green, red, yellow, brown, or purple? Although it is true that each of these hues can be measured by instruments to determine its wavelength, and the number assigned to one hue would be greater or less than the number assigned to another, we don't think in these terms when we perceive hue. We perceive different hues only as categorically different, not quantitatively different; one hue is not more or less than another, they're just different. In contrast, we perceive color intensity quantitatively, from low to high.

Whether we perceive visual attributes as quantitative or categorical is significant. How we perceive these attributes tells us which ones we can use to encode

3-D position is also a preattentive attribute, but it cannot be applied effectively to graphs on a flat surface (e.g., a page or computer screen), which relies on illusory cues of light and shadow, occlusion, and size to simulate depth. Even if 3-D position could be used effectively on a flat surface, it is not an effective attribute to use in table and graph design because our perception of depth is weak compared to our perception of height and width.

quantitative values in graphs and which can only be used to encode categorical differences. The following list indicates which preattentive attributes are perceived quantitatively:

Type	Attribute	Quantitatively Perceived?
Form	Length	Yes
	Width	Yes, but limited
	Orientation	No
	Size	Yes, but limited
	Shape	No
	Enclosure	No
Color	Hue	No
	Intensity	Yes, but limited
Position	2-D position	Yes

Although one might argue that we can also perceive orientation quantitatively, there is no natural association of greater or lesser value with this characteristic (e.g., which is greater, a vertical or a horizontal line?).

All of these preattentive attributes can be used to encode categorical differences, including those that we perceive quantitatively. The most effective of these is 2-D position. For instance, the 2-D position of data points in a graph can simultaneously represent quantitative values based on the position of the data point relative to the Y axis and categorical differences based on their position along the X axis, as in the following example:

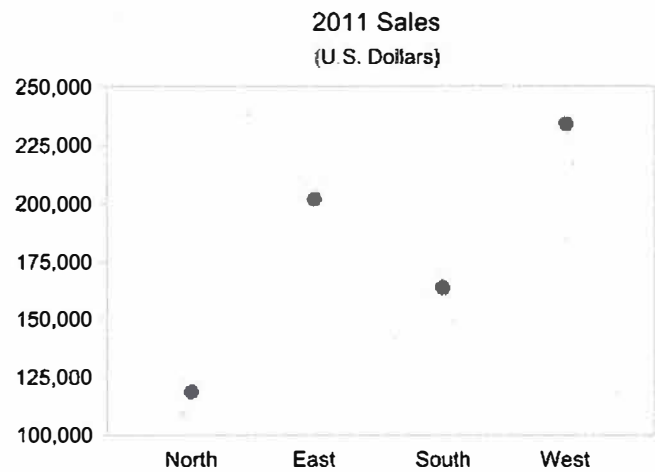


FIGURE 5.12 This graph uses the 2-D position of the data points to encode quantitative values along the Y axis and categorical items along the X axis.

Whenever you can use either length or 2-D position to encode quantitative values in graphs, you should do so rather than using other attributes that can be perceived quantitatively only to a limited degree. Even though we can tell that one value is greater than another when we use width, size, or color intensity, these attributes do not indicate precisely how much a value differs. This is true of width and size because both rely on our ability to assign values to the 2-D areas of objects, such as the slices of a pie chart or the width of a line. Although we can do a fairly good job of discerning that one object has a larger area than another, it is difficult to perceive by how much or to assign a value to the area. Take a look at the two circles on the following page. I've assigned a value to the one on the left, and it is up to you to assign a value to the one on the right.

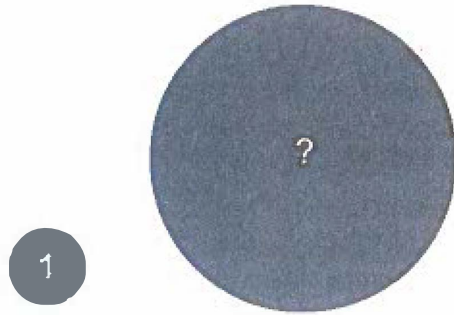


FIGURE 5.13 This figure illustrates the difficulty of perceiving differences in the 2-D areas of objects.

How much bigger is the large circle than the small circle? Given the fact that the size of the small circle equals a value of 1, what is the size of the large circle?

* * * * *

The answer is 16. If you got the answer right, either you got lucky or you have rare perceptual talent. When I ask this question in a class of 70 students, answers typically range from 5 to 50.

Color intensity presents a similar problem. For instance, we can use color intensity along a gray scale ranging from white as the minimum value to black as the maximum. As long as there is enough difference in the color intensity of two objects, we can tell that one is darker than the other, but it is hard to assign a value to that difference. For this reason, it is best to avoid color intensity as a means of encoding quantitative value if the attributes of length or 2-D position are available to use instead.

Perceptual Effects of Context

One fact that may surprise you about our perception of visual attributes such as color or size is that our visual senses are not designed to perceive absolute values but rather differences in values. Your sense of the brightness of a candle will differ depending on the context, i.e., in relation to the other light in your range of vision. If you light a candle when surrounded by bright sunshine, the candle won't seem very bright. If you light the same candle while standing in a pitch-black cellar, the candle will seem extremely bright in contrast to the darkness. That's because our visual receptors measure differences rather than absolute values. Look at the four squares contained in the large rectangle below:

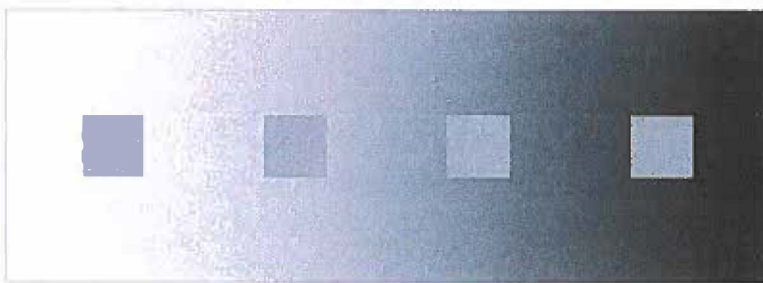
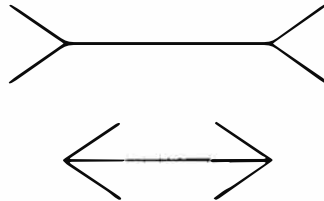


FIGURE 5.14 This illustration demonstrates the effect of context on the perception of color intensity.

How much darker is the square on the left than the square on the right?

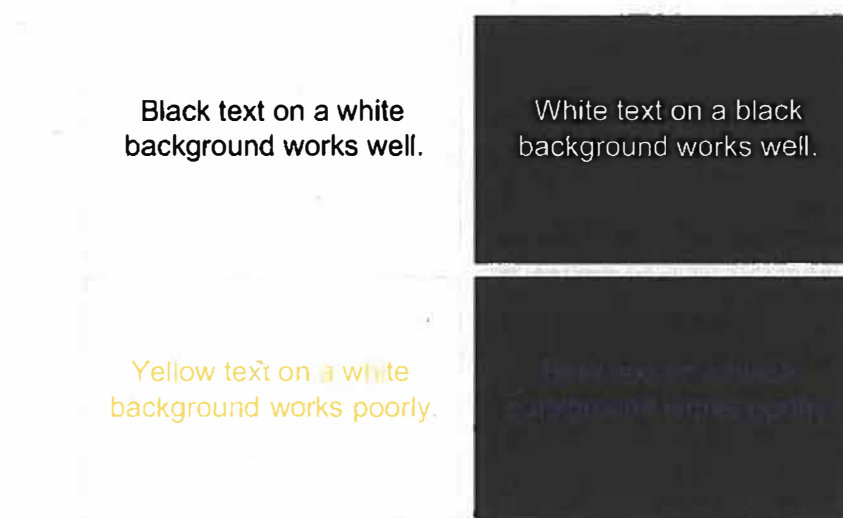
The truth is, there is no difference at all in the actual values of the four squares when measured by an instrument. All four are 50% gray, precisely in the middle between pure white and pure black. They appear different to us in this illustration, however, because our perception of them is influenced by the differences in the color intensity that surrounds them.

Every attribute of visual perception is influenced by context. In the following figure, the upper horizontal line looks longer than the lower, but they are exactly the same. The different shapes at the ends of the two horizontal lines affect our perception of their lengths.



Optical illusions such as these clearly demonstrate the effect of context on visual perception. Even when we know we are looking at an optical illusion, misperception persists, because visual perception measures differences in the attributes of what we see, not absolute values. It's up to us to use this knowledge when designing tables and graphs to prevent context from causing a misperception of the message we're trying to communicate.

Some combinations of an object's visual attributes and its context can work well to highlight the object, but some can render it imperceptible. For instance, black text on a white background is easy to read. Other combinations are also acceptable, such as white text on a black background, and, to a lesser degree, red or dark blue text on a white background. In contrast, some combinations simply don't work, such as yellow on a white background or blue on a black background.



We could examine many more examples, both good and bad, but it's not necessary to memorize the combinations that do and don't work. As you can see

One practical application of the lesson that we learn from *Figure 5.14* is that we shouldn't use a gradient of fill color in the background of a graph. Doing so would cause objects (e.g., bars) on the left to look different from objects on the right even if they're exactly the same.

FIGURE 5.15 This figure demonstrates the effect of context on the perception of length.

FIGURE 5.16 Some contrasting hues work well for clear perception, and some don't. You must look closely to see that there is any text at all in the lower right rectangle because blue does not stand out distinctly against a black background.

from these examples, it's fairly easy to tell the difference when you take the time to look. Pay attention and use your own sense of visual perception to select combinations that are effective for clear and efficient communication and avoid those that aren't.

Limits to Distinct Perceptions

There is a limit to the number of expressions of a single attribute that can be clearly distinguished in a graph. For instance, if you create a line graph displaying departmental expenses across time, with a separate line of a distinct hue to represent each department in your company, readers of the graph will only be able to easily distinguish a limited number of hues. This is true for each visual attribute that can be used to encode categorical items.

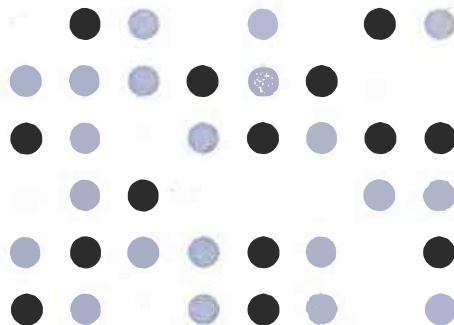
Our ability to distinguish visual attributes diminishes as the number of alternatives increases. According to Ware:

Pre-attentive symbols become less distinct as the variety of distracters increases. It is easy to spot a single hawk in a sky full of pigeons, but if the sky contains a greater variety of birds, the hawk will be more difficult to see. A number of studies have shown that the immediacy of any pre-attentive cue declines as the variety of alternative patterns increases, even if all the distracting patterns are individually distinct from the target.³

According to research, when reading graphs, we can distinguish preattentively between no more than about eight different hues, about four different orientations, and about four different sizes. All the other visual attributes of preattentive perception should be limited to a few values as well.⁴ When we introduce a greater number of alternatives, people can no longer distinguish them preattentively and must therefore switch to slower attentive processing.

You might think that it would be worthwhile to use a larger number of distinct values even though this would force your readers make use of slower, attentive processing to interpret your graph. Unfortunately, if you do this, your readers will run up against the limits of working memory, which allows them to remember no more than four distinct values at a time.

Another limitation to keep in mind is that preattentive processing usually cannot handle more than one visual attribute of an object at a time. Let me illustrate. In the figure below, only the attribute of color intensity is used to distinguish between three values, represented with black, dark gray, and light gray:



3. Colin Ware (2004) *Information Visualization: Perception for Design*, Second Edition. Morgan Kaufmann Publishers, page 152.

4. Ibid. page 182

FIGURE 5.17 This figure illustrates the use of distinct values of a single visual attribute, in this case color intensity, to encode objects as different from one another.

It is not difficult to pick out the black circles as distinct from those that are dark gray or light gray, the dark gray circles as distinct from those that are black or light gray, and so on. Let's complicate it now by introducing a second preattentive attribute. By using two shapes (circles and squares), along with the three color intensities (black, dark gray, and light gray), we double the number of distinct values, but we do so at a cost. Take a look.

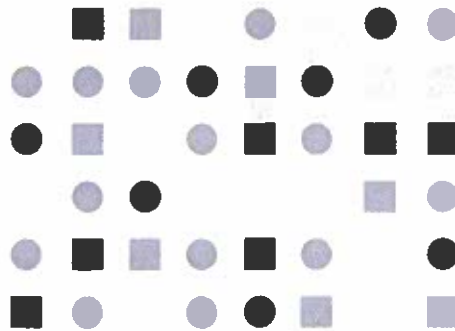


FIGURE 5.18 This figure illustrates the problem that results from encoding distinct values using more than one visual attribute, in this case color intensity and shape.

It's still fairly easy to focus on just the squares, or just the circles, or just the black objects, or just the dark gray objects, or just the light gray objects, but try to pick out the dark gray squares or the light gray circles. The process of simultaneously focusing on multiple visual attributes (e.g., shape and color intensity) requires slower, attentive processing. You can imagine how much more complicated it would get if we threw in a few more shapes or added the third attribute of size.

A related consideration is that when you select the expressions of a single attribute that you will use in a graph (e.g., multiple hues to distinguish categorical items), you must make sure that they are far enough apart from each other along the continuum of potential expressions to stand out clearly as distinct. Although I can choose up to four different shades of gray, shades that are too close to one another won't work. Here are two sets of four distinct values, encoded as different shades of gray. The four shades on the left were carefully chosen from the full range of potential values so they could easily be distinguished from one another, but those on the right were not.

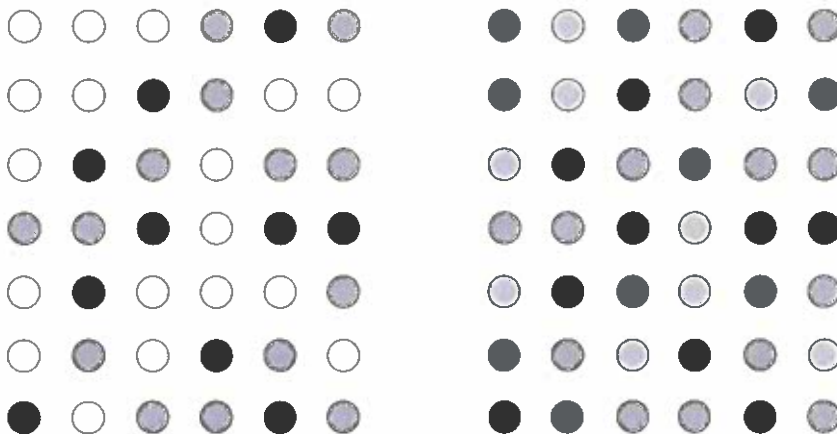


FIGURE 5.19 This figure illustrates the importance of selecting values of a visual attribute that can be easily distinguished. The four shades of gray encoded in the circles on the left are distinct, but the four on the right are not.

With a little effort, you can develop once and for all a set of distinct values for each of the visual attributes, and you can use this set over and over, rather than repeatedly going through the process of determining the distinctions that work.

Here's a list of nine hues that are easy to recognize and distinct enough to work well together:

1. Gray
2. Blue
3. Orange
4. Green
5. Pink
6. Brown
7. Purple
8. Yellow
9. Red

Each of these hues meets the requirement of distinctness, but the context of their use should sometimes lead you to select some and reject others. A good example, especially if you communicate with an international audience, involves cultural context. Particular hues carry meanings that vary significantly among cultures. In most western cultures, we think of red as signifying danger, warning, heat, and so on, but in China, red represents good fortune. The key is to be in touch with your audience and take the time to consider any possible associations with particular hues that might influence their interpretation of your message.

Various colors also affect us in different ways on a fundamental psycho-physical level. Some colors are strong and exciting, grabbing our attention, and others are more neutral and soothing, fading more into the background of our attention. Edward Tufte suggests that colors found predominantly in nature, especially those that are not overly vibrant (e.g., medium shades of gray, green, blue, and brown), are soothing and easy on our eyes.⁵ Such colors are particularly useful in tables and graphs for anything that you don't want to stand out above the other content. On the other hand, fully saturated, bright versions of just about any hue will demand attention. These should only be used when you want to highlight particular information. Notice the difference between the medium shades on the left and the brighter shades on the right:



An excellent source of color combinations that work well together for data presentation can be found on the website www.ColorBrewer.org.

5. Edward R. Tufte (1990) *Envisioning Information*. Graphics Press, page 90.

FIGURE 5.20 This figure shows soft, soothing hues on the left, and bright, attention-getting hues on the right.

Another consideration when selecting a palette of colors is that our ability to distinguish colors decreases along with the sizes of objects. For this reason, small objects such as data points in graphs should be darker than large, visually weighty objects such as bars. In the two graphs below, the same colors are used in both. The colors of the bars in the upper graph are quite distinct, but they are somewhat difficult to distinguish in the lower graph because the points are small.



FIGURE 5.21 The same colors that allow us to easily distinguish bars are much harder to distinguish when used for small objects such as data points.

When selecting your palettes of colors for encoding categorical distinctions in graphs, it works best to maintain three versions: one with relatively light, yet distinct colors for large objects such as bars, one with darker versions of the same colors for small objects such as lines, and one with bright versions of the same colors for objects that must stand out above the rest. Here's an example of three versions of the nine distinct hues that were listed previously, two versions for ordinary use and one for highlighting:

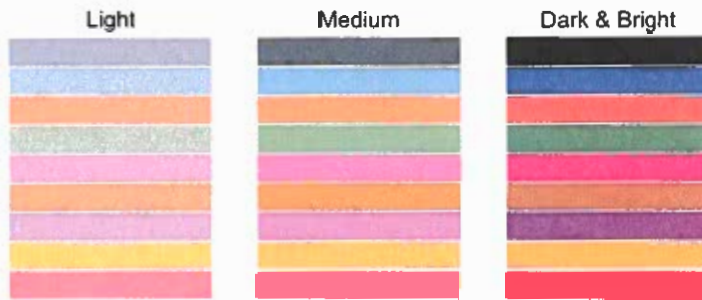


FIGURE 5.22 These sample colors could be used for the three color palettes that you should maintain for displaying data. The RGB values of each are provided in Appendix I – *Useful Color Palettes*.

Another consideration when using hues to encode data is that about 10% of males and 1% of females suffer from colorblindness, unable to distinguish certain colors because of a lack of cones (color receptors) in the eye that detect a certain range of hue. Most people who are colorblind lack the cones that enable them to distinguish between red and green so that, to these viewers, both colors appear brown. When your audience is likely to include people who are colorblind, it is best to use only red or green, but not both, unless you are careful to vary their intensities sufficiently so that they appear different from one another.

Limits to the Use of Contrast

Visual perception evolved in a way that makes us particularly aware of differences, i.e., features that stand out in contrast to the rest. When we look at something, our eyes are not only drawn to differences, but our brains insist on making some sense of those differences. If you create a table of information and all the text is black except for one column that is red, readers will assume that the difference is significant and that the red text is especially important. It is up to us to only include visual differences that are meaningful.

Contrast is most effective when only one thing is different in a context of other things that are the same. As the number of differences increases, the degree to which those differences stand out decreases. In the graph below, the attributes were controlled so that the bar representing “Contributions from People like You” would clearly stand out from the rest because of a single difference: its color.

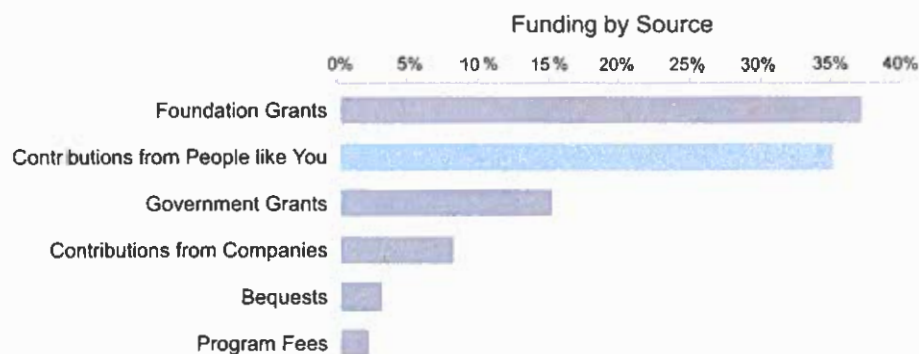


FIGURE 5.23 This graph uses contrast effectively to highlight the most important information.

If you try to emphasize so many things that emphasis becomes the rule rather than the exception, your message will become buried in visual clutter. When everyone in the room shouts, no one can be heard.

Gestalt Principles of Visual Perception

Many fields of scientific study have contributed to our understanding of visual perception, but none have revealed more that is relevant to visual design than the Gestalt School of Psychology. The original intent of Gestalt research when it began in 1912 was to uncover how we perceive pattern, form, and organization in what we see. The German word *gestalt*, however lofty it may seem to speakers of English, simply means “pattern.” The founders observed that we organize what we see in particular ways in an effort to make sense of it. The result of their research was a series of Gestalt principles of perception, which are still respected today as accurate descriptions of visual behavior. These principles reveal that we tend to group objects in particular ways. Many of these findings are relevant to our interest in the design of tables and graphs as well as to the overall design of the reports that contain them. This section is an introduction to Gestalt principles.

Principle of Proximity

We perceive objects that are close to each other as belonging to a group. In the following illustration, we naturally perceive the 10 circles as three groups because of the way they are spatially arranged:

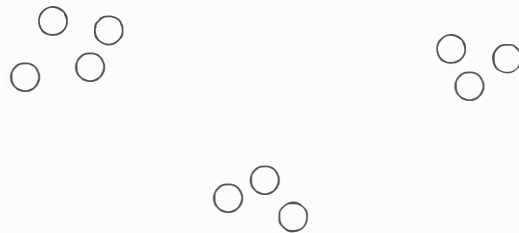


FIGURE 5.24 This figure illustrates the Gestalt principle of proximity.

This principle can be applied to the design of tables. We can direct our readers to scan predominantly across rows or down columns of data by spacing the data to accentuate either the row or the column groupings, as illustrated below:

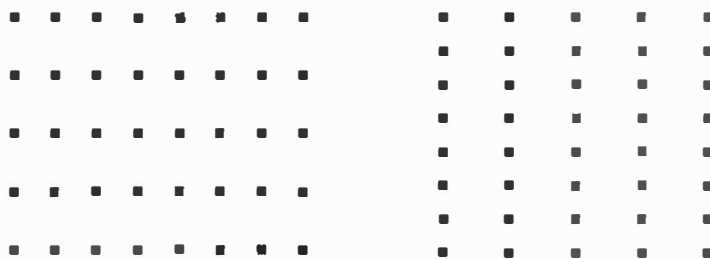


FIGURE 5.25 This figure illustrates an application of the Gestalt principle of proximity to table design

In response to subtle grouping cues, we are naturally inclined to scan the squares on the left by row and those on the right by column. This is because the squares on the left are slightly closer together horizontally than vertically, which groups them by rows, and the squares on the right are slightly closer together vertically than horizontally, which groups them by columns. The conscious use of space to group objects together or keep them separate is a powerful tool for organizing information and directing your readers to view it in a particular way.

Principle of Similarity

We tend to group together objects that are similar in color, size, shape, or orientation. Here are some examples:

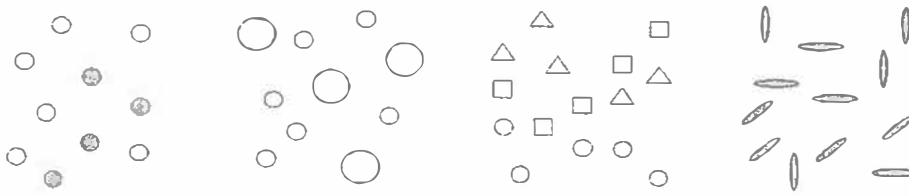


FIGURE 5.26 These examples illustrate the Gestalt principle of similarity.

This principle reinforces what we've already learned about the usefulness of color (both hue and intensity), size, shape, and orientation to display categorical groupings. When these attributes are used to encode a distinction between different sets of data in a graph, they work quite effectively as long as the number of distinctions is kept to a few and the attributes are different enough from one another so that we can clearly distinguish the groups. We can use this principle to direct readers of tables to focus primarily across rows or down columns, as in the following example:

XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX
XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX
XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX
XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX
XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX
XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX
XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX
XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX
XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX	XXXXX

FIGURE 5.27 This figure demonstrates the use of the principle of similarity to arrange data column-wise in a table.

Principle of Enclosure

We perceive objects as belonging together when they are enclosed in a way that appears to create a boundary around them (e.g., a border or common field of color). They appear to be set apart in a region that is distinct from the rest of what we see. Notice how strongly your eyes are led to group the enclosed objects below:

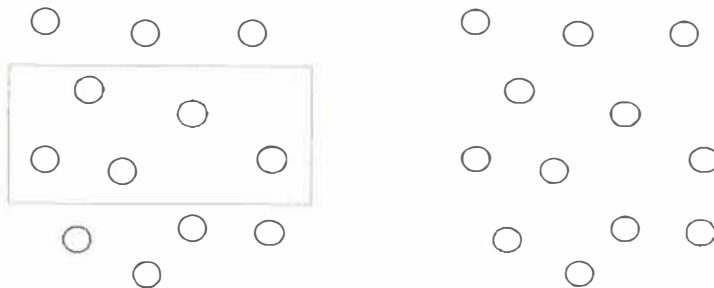


FIGURE 5.28 These two examples illustrate the Gestalt principle of enclosure; in the left-hand example, the enclosure is formed by a border, and, in the right-hand example, by a gray region.

The two sets of circles are arranged in precisely the same way, yet the differing enclosures lead us to group the circles differently. This principle is exhibited frequently in tables and graphs by the use of borders and background fill colors to group information and sometimes to emphasize particular items. As you can see in the previous illustration, a strong enclosure (e.g., bright, thick lines or dominant colors) is not necessary to create a strong perception of grouping. In fact, we sometimes perceive enclosures that aren't really there, such as the white shape that you see in the middle of the gray objects below:

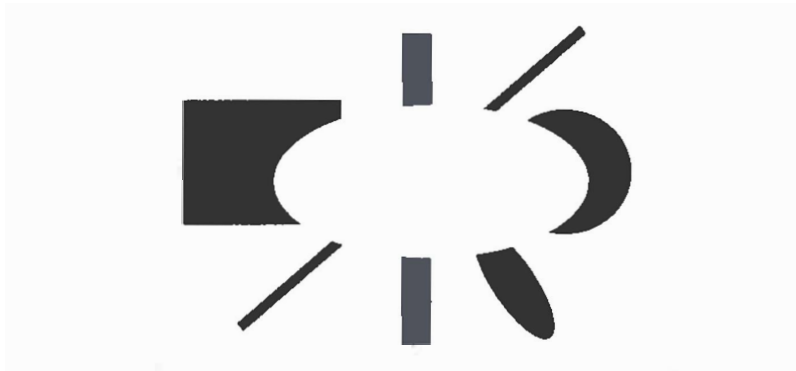


FIGURE 5.29 This is an example of an illusory contour (i.e., perception of a white oval in the middle of the gray objects, forming a separate region of white space).

The reason that we perceive this illusory enclosure is explained by the next two Gestalt principles: the principle of closure and the principle of continuity.

Principle of Closure

We have a keen dislike for loose ends. We will always try to resolve ambiguous visual stimuli to eliminate the ambiguity. So if we are faced with objects that could be perceived on the one hand as open, incomplete, and unusual forms; or on the other hand as closed, whole, and regular forms, we will naturally perceive them as the latter. In the illusory enclosure illustrated above, it is natural to perceive the gray objects as rectangles, circles, ovals, and continuing lines, partially hidden by a white oval, rather than as a collection of oddly shaped objects. The principle of closure asserts that we perceive open structures as closed, complete, and regular (e.g., the gray objects above) whenever there is a way that we can reasonably interpret them as such. Here's a more direct illustration of this principle:

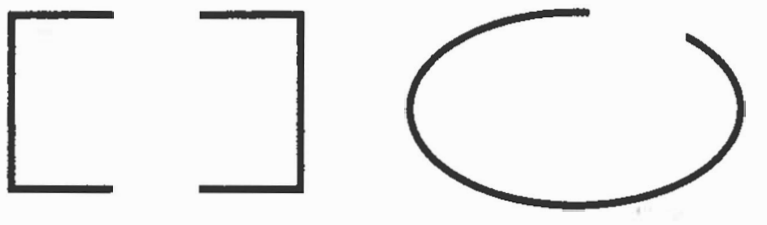


FIGURE 5.30 This figure illustrates the principle of closure.

It is natural for us to perceive the lines on the left as a rectangle rather than as two sets of three connected and perpendicular lines and to perceive the object on the right as an oval rather than a curved line.

We can put this tendency to perceive whole structures to use when designing tables and graphs. For example, we can group objects (e.g., points, lines, or bars

on a graph) into visual regions without the use of complete borders to define the space. It is not only sufficient but usually preferable to define the area of a graph through the use of a single set of subtle X and Y axes rather than with heavy lines and fill colors that enclose it completely, as in the following examples:

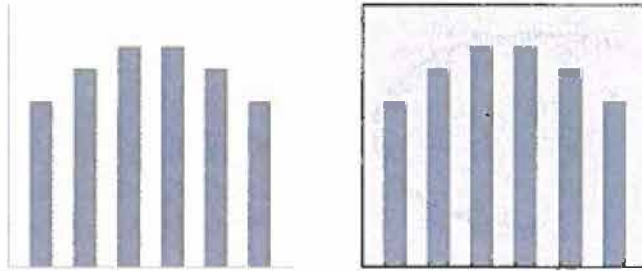


FIGURE 5.31 These examples demonstrate an application of the principle of closure. We can define the area of a graph with subtle X and Y axes, as seen on the left, thus minimizing the visual weight of a graph's supporting components in relation to its data components.

Principle of Continuity

We perceive objects as belonging together, as part of a single whole, if they are aligned with one another or appear to form a continuation of one another. In the left-hand illustration below, we see the various visual objects as forming a simple image of a rectangle and a wavy line. If we separated the two objects, we assume they would look like those in the middle illustration. We don't see a rectangle and three curved lines, like those in the right-hand illustration even though this is another possible interpretation of the image. Our tendency is to see the shapes as continuous to the greatest degree possible.

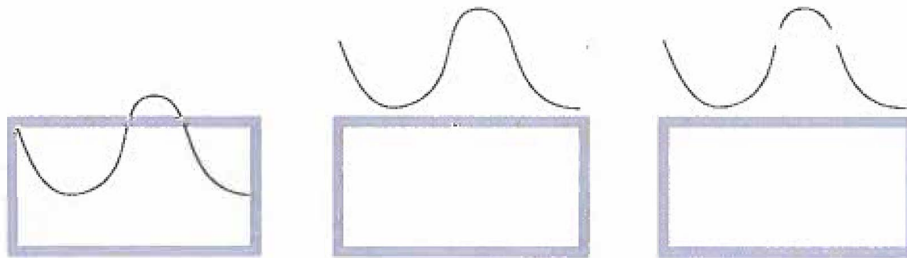


FIGURE 5.32 These examples illustrate the principle of continuity.

In the bar graph below, the left alignment of the bars makes it obvious that they share the same baseline, thus eliminating the need to reinforce the fact with a vertical axis:

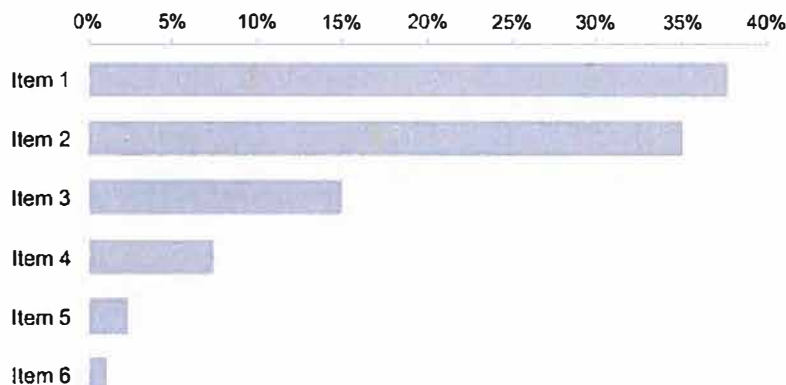


FIGURE 5.33 This is an example of applying the principle of continuity to graph design.

In the table below, it is obvious which categorical items are division names and which are department names, based on their distinct alignments:

Division	Department	Headcount
G&A	Finance	15
	Purchasing	5
	Information Systems	17
Sales	Field Sales	47
	Sales Operations	10
Engineering	Product Development	22
	Product Marketing	5

FIGURE 5.34 This is an example of applying the principle of continuity to table design.

The left alignment of the divisions and departments and the right alignment of the headcounts creates a strong sense of grouping without any need for vertical grid lines to delineate the columns. Even if two columns of data overlap, with no white space between them, as long as they are separately aligned, they remain clearly distinguishable, as in the following example:

Division/Department	Headcount
G&A	
Finance	15
Purchasing	5
Information Systems	17
Sales	
Field Sales	47
Sales Operations	10
Engineering	
Product Development	22
Product Marketing	5

FIGURE 5.35 This is an example of applying the principle of continuity to table design to save horizontal space.

In this example, the principle of continuity was used to align related categories in a way that saved horizontal space.

Principle of Connection

We perceive objects that are connected (e.g., by a line) as part of the same group. In the example below, even though the circles are equally spaced from one another, the lines that connect them create a clear perception of two vertically attached pairs:



Technically, the principle of connection was not directly presented by the original Gestalt School of Psychology but has been articulated since as an extension of the principle of continuity.

FIGURE 5.36 This figure illustrates the principle of connection.

Connection exercises even greater power over visual perception than proximity or similarity (color, size, and shape) but less than enclosure, as you can see from the examples below:



FIGURE 5.37 These are examples of the relative perceptual strength of the Gestalt principles that we’ve examined so far.

The perceptual strength of connection makes lines a useful method of connecting items in graphs. Points without lines in graphs are much harder for our eyes to connect. Notice the difference between the two examples below:

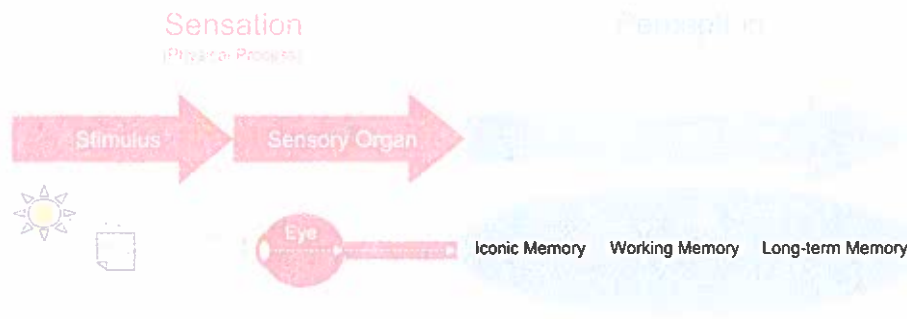


FIGURE 5.38 These examples illustrate the power of lines to reveal connectedness.

Lines in a graph not only create a clear sense of connection but also bring to light the overall shape of the data, when appropriate, which couldn't be discerned without them.

Summary at a Glance

Mechanics of Sight



Attributes of Preattentive Processing

Type	Attribute	Quantitatively Perceived?
Form	Length	Yes
	Width	Yes, but limited
	Orientation	No
	Size	Yes, but limited
	Shape	No
	Enclosure	No
	Hue	No
Color	Intensity	Yes, but limited
	2-D position	Yes

Gestalt Principles of Visual Perception

Principle	Description
Proximity	Objects that are close together are perceived as a group.
Similarity	Objects that share similar attributes (e.g., color or shape) are perceived as a group.
Enclosure	Objects that appear to have a boundary around them (e.g., formed by a border or area of common color) are perceived as a group.
Closure	Open structures are perceived as closed, complete, and regular whenever there is a way that they can be reasonably interpreted as such.
Continuity	Objects that are aligned together or appear to be a continuation of one another are perceived as a group.
Connection	Objects that are connected (e.g., by a line) are perceived as a group.